

FRictional Force

If two systems are **in contact and stationary** relative to one another, then the friction between them is called **static friction**.

If two systems are in contact and moving relative to one another, then the friction between them is called **kinetic friction**.

Formula of Frictional Force

$$F_{frict} = \mu \cdot F_{norm}$$

Finding the Normal Force:

$$N = mg$$

- In this formula, m describes the object's mass, and g stands for the acceleration due to gravity.

In the case of an **inclined surface**, the strength of the normal surface is reduced the more the surface is inclined, hence the formula becomes:

$$N = mg \cos \theta$$

The co-efficient you choose **depends on the object** and the specific situation. If the object isn't moving across the surface, you use the coefficient of static friction, but if the object under consideration is moving, you use the coefficient of sliding friction.

The type of materials used also affects the co-efficient. For example, if a block was on a brick surface, the coefficient would be 0.6, but if it were on a block of clean wood, it would range from 0.25 to 0.5.

SYSTEM	STATIC FRICTION (μ_s)	KINETIC FRICTION (μ_k)
Rubber on dry concrete	1.0	0.7
Rubber on wet concrete	0.5-0.7	0.3-0.5
Wood on wood	0.5	0.3
Waxed wood on wet snow	0.14	0.1
Metal on wood	0.5	0.3
Steel on steel (dry)	0.6	0.3

Steel on steel (oiled)	0.05	0.03
Teflon on steel	0.04	0.04
Bone lubricated by synovial fluid	0.016	0.015
Shoes on wood	0.9	0.7
Shoes on ice	0.1	0.05
Ice on ice	0.1	0.03
Steel on ice	0.04	0.02

EXAMPLE PROBLEM/S

1. A large block of ice is being pulled across a frozen lake. The block of ice has a mass of 300 kg. The coefficient of friction between two ice surfaces is small: $\mu_k = 0.05$. What is the force of friction that is acting on the block of ice?

Given:

$$m = 300 \text{ kg} \quad g = 9.8 \text{ m/s}^2$$

$$\mu_k = 0.05$$

Solution:

$$F_f = \mu N$$

$$F_f = \mu mg$$

Substituting the values we get:

$$F_f = 0.05 \times 300 \text{ kg} \times 9.8 \text{ m/s}^2$$

We get: 147 kg m/s^2 or **147 N**.

$$\boxed{F_f = 147 \text{ N}}$$

2. A man has to push his boat on the shore across the mud to get to the water. The coefficient of friction between the boat and the mud is given by $\mu = 0.400$. If the boat has a mass of 40 kg, calculate the magnitude of the force of friction acting on the boat.

Given:

$$m = 40 \text{ kg} \quad g = 9.8 \text{ m/s}^2$$

$$\mu_k = 0.4$$

Solution:

$$F_f = \mu N = \mu mg$$

$$F_f = (0.4)(40 \text{ kg})(9.8 \frac{\text{m}}{\text{s}^2})$$

$$\boxed{F_f = 156.8 \text{ N}}$$